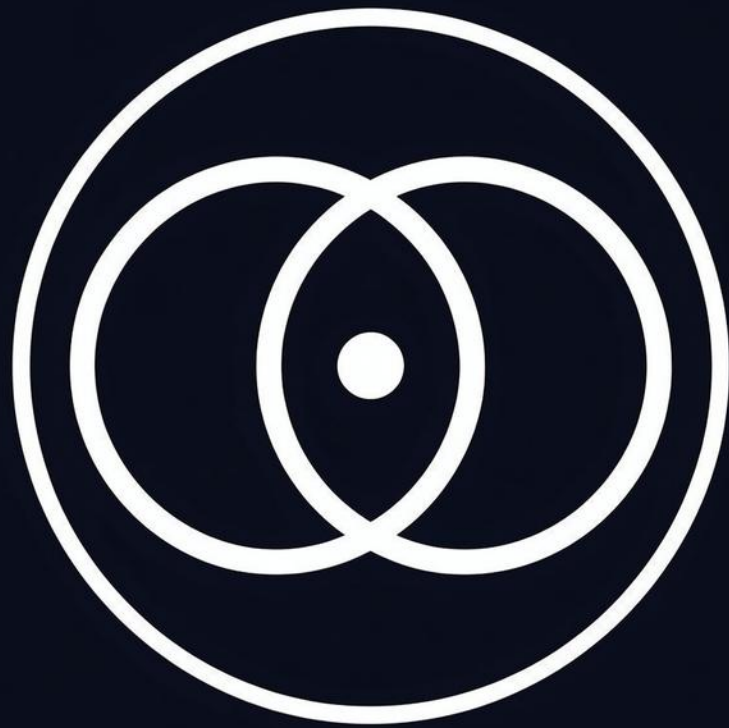




P0c

元认知：知之为知之，不知为不知——A I 自我认知与边界校准的工程化

Metacognition: Know What You Know and What You Do Not —
Engineering AI Self-Knowledge and Boundary Calibration

TwinsEarth · 智能体宇宙论文系列

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元认知：知之为知之，不知为不知——AI 自我认知与边界校准的工程化

Metacognition: Know What You Know and What You Do Not — Engineering AI Self-Knowledge and Boundary Calibration

Working Title (EN): Engineering AI Self-Knowledge: Calibration, Abstention, and Identifiability

旗舰三部曲序章 P0c • 与 P0《空间结构，本身就是智能》、P0b《结构+规模，本身就是智能》构成"结构 × 规模 × 自我边界"三轴 • UDOS 推演引擎 v7.5.0 • 2026-09-20

一句话先说清：孔子说"知之为知之，不知为不知，是知也"。今天的 AI 恰恰做不到这一点——它知识量近乎无限，却不知道自己的答案边界在哪里。本文不讨论 AI 有没有意识，只回答一个工程问题：**怎样让一台机器在它不知道的时候，诚实地说"我不知道"。**

证据口径（全文只声明这一次）：文中 UDOS 引擎的数字全部来自 reports7/*.json，普通 CPU、固定随机种子、可重复运行，分 verified（有测试守门）、cpu-proto/cpu-proxy（CPU 原型趋势）、unverified（外部动机）三级。凡单种子试点均写明，投稿前需 ≥30 个种子复算。文献逐条核验于 2026-09-20。

Prologue P0c of the flagship trilogy • together with P0 (Space Structure Is Itself Intelligence) and P0b (Structure + Scale Is Itself Intelligence) it forms the three axes "structure × scale × self-boundary" • UDOS reasoning engine v7.5.0 • 2026-09-20

The thesis in one sentence. Confucius said: "To know what you know and what you do not know, that is true knowledge." Today's AI fails exactly here — its knowledge is nearly unbounded, yet it does not know where its own answers stop being reliable. This paper does not ask whether AI is conscious. It asks one engineering question: **how do we make a machine honestly say "I don't know" when it doesn't know?**

Evidence policy (stated once). Every UDOS number below comes from reports7/*.json, produced on an ordinary CPU with fixed random seeds and fully reproducible runs. Levels: verified (guarded by automated tests), cpu-proto/cpu-proxy (trends observed on a CPU prototype), unverified (external motivation only). Single-seed pilots are labeled as such; ≥30 seeds are required before submission. References were checked individually on 2026-09-20.

摘要

Abstract

当前最强的大模型，本质上是一个超级概率匹配器：给它一个问题，它在海量参数里找出统计上最像答案的那个，然后用流畅的语言输出。这套机制有一个致命缺口——**它没有"我不知道"这个概念。**遇到训练数据里没有的问题，它不会停下来承认边界，而是自信地生成一个看似合理的错误答案。这就是幻觉的根源：不是知识不够，而是缺少对自身知识边界的感知（元认知）。

The strongest language models today are, at bottom, supercharged probability matchers: given a question, the model finds the statistically most likely continuation in a vast parameter space and renders it in fluent prose. This mechanism has one fatal gap — **it has no concept of "I don't know."** Confronted with a question outside its training data, it does not pause and acknowledge a boundary; it confidently produces a plausible-looking wrong answer. That is the root of hallucination: not a lack of knowledge, but a lack of metacognition — a sense of the boundaries of one's own knowledge.

本文刻意不碰"AI 是否有意识"这个形而上学问题，而是把"自知"降维成三个可以直接测量、可以验收、可以证伪的工程量：

This paper deliberately sets aside the metaphysical question of machine consciousness. Instead it reduces "self-knowledge" to three quantities that can be directly measured, acceptance-tested, and falsified:

1. **校准**：模型说"我有八成把握"时，是否真的有八成正确率；
2. **选择性拒答**：在高风险样本上主动弃权，用"风险—覆盖率"曲线衡量拒得准不准；
3. **可辨识性自知**：知道哪些隐变量在当前观测条件下根本无法估计，从而拒绝硬猜。

1. **Calibration**: when the model says "I am 80% confident," is it actually right 80% of the time?
2. **Selective abstention**: declining to answer on high-risk inputs, with the quality of refusal judged by a risk-coverage curve.
3. **Identifiability awareness**: knowing which latent variables cannot, in principle, be estimated under the current observation setup, and refusing to guess them.

我们在一个 967,796 参数的小型世界模型上给出三组直接证据。其一，分裂共形预测在 80%/90%/95% 三个置信水平上全部通过覆盖率检验（实测 95% 水平覆盖 0.9577），即使对"看不到隐藏参数"的盲路径也成立。其二，在一个 6 帧短窗口里估计加速度，估计器的误差比"直接报平均值"还差 50.8%（自知分数 -0.508）——正确动作是弃权或加长观测，而不是硬估。其三，21 项门控检查全部通过，接触类任务在没有动作先验时成功率为 0%，引入门控后升到 100%。

We report three sets of direct evidence from a small world model of 967,796 parameters. First, split conformal prediction passes coverage at all three nominal levels, 80%/90%/95% (observed 0.9577 at the 95% level), including a "blind" path that cannot see the latent parameters. Second, estimating acceleration from a short 6-frame window yields an estimator whose error is 50.8% worse than simply predicting the mean (skill score -0.508) — the correct action is to abstain or lengthen the observation window, not to force an estimate. Third, all 21 release gates pass; on contact tasks, direct control without a motion prior succeeds 0% of the time, rising to 100% once a gate routes those tasks to a hybrid policy.

这些都只是 CPU 原型上的单种子初步证据，不是已在真实大模型上验证的定律。本文的真正贡献是一个研究纲领：把"智慧"这个无法直接测量的词，替换成三个可测量的工程必要条件，并给出五条可以被实验推翻的预测。在这三件事被做成可测量、可验收之前，任何"系统有智慧"的宣称都不严谨。

These are single-seed preliminary results on a CPU prototype, not laws validated on real large models. The real contribution is a research program: replacing the unmeasurable word "wisdom" with three measurable engineering prerequisites, together with five experimentally falsifiable predictions. Until these three things are measurable, acceptance-tested, and falsifiable, any claim that a system "has wisdom" is not rigorous.

关键词：元认知；幻觉；校准；选择性预测；共形预测；结构可辨识度；世界模型；安全门控

Keywords: metacognition; hallucination; calibration; selective prediction; conformal prediction; structural identifiability; world models; safety gating

1 一辆马力巨大、却没有刹车的车

1 A Car with a Huge Engine and No Brakes

1.1 孔子的体检报告

1.1 Confucius as a diagnostic report

《论语·为政》里有一句被引用了两千五百年的话：“**知之为知之，不知为不知，是知也。**” 它讲的不是知识量，而是对自身知识状态的觉察：知道自己知道什么，是清醒；知道自己不知道什么，是智慧。

In the Analects (II.17), Confucius says: "**To know what you know and what you do not know, that is true knowledge.**" The sentence is not about the amount of knowledge; it is about awareness of one's own epistemic state: knowing what you know is clarity; knowing what you do not know is wisdom.

用这句话给今天的 AI 做一次体检，结果很有意思。论知识量，一个大模型读过的文本超过任何人类一生所能接触的；但论自我觉察，它停留在婴儿阶段——它不知道自己的答案是从哪来的、可信度多少，更不知道自己的知识边界在哪里。

Run today's AI through that diagnostic and the result is striking. In sheer volume of knowledge, a large model has read more text than any human could encounter in a lifetime. In self-awareness it remains an infant: it does not know where its answers come from, how reliable they are, or where its knowledge ends.

需要先立一条纪律：**孔子这句话是一个道德理想（应该如此），不是科学证据（实际如此）**。本文引用它，只是为了说明我们想工程化的那个古老直觉是什么；本文的全部科学主张都建立在第 2 章的三个可测量量和第 4 章的实测上。如果本文的预测被实验推翻，孔子也救不了它。

One rule must be set first: **Confucius's sentence is a moral ideal (an ought), not scientific evidence (an is).** We cite it only to name the ancient intuition we are trying to engineer. Every scientific claim in this paper rests on the three measurable quantities of Section 2 and the experiments of Section 4. If our predictions are experimentally refuted, Confucius cannot save the paper.

1.2 幻觉的精确定义

1.2 A precise definition of hallucination

人们常把幻觉理解为“模型说错了”。这个定义太宽——任何有限的模型都会犯错。本文采用一个更精确、也更危险的定义：

Hallucination is often defined loosely as "the model says something wrong." That definition is too broad — every finite model errs. We adopt a sharper and more dangerous definition:

幻觉，是模型对一个它本质上不知道的东西，输出了一个与正确答案同样自信的回答。

Hallucination is when a model outputs, for something it fundamentally does not know, an answer every bit as confident as its answer to something it does know.

前者（答错）是能力不足；后者（自信地答错）是自我边界缺失。一个有自知的系统，在能力不足时应该输出宽区间、弃权，或转去查资料；一个没有自知的系统，在能力不足时仍然给出笃定的点估计。

The first (being wrong) is a competence deficit; the second (being confidently wrong) is a missing self-boundary. A self-aware system, when out of its depth, should emit a wide interval, abstain, or go retrieve information; a system without self-awareness still hands back a firm point estimate.

为什么概率匹配器天然缺少边界感？因为它的训练目标是"预测下一个 token"，这个目标里根本没有"我是否知道"这一维。它被训练成永远给出一个最可能的续写，而不是评估自己该不该续写。

Why does a probability matcher lack a boundary by construction? Because its training objective — predict the next token — contains no dimension for "do I actually know this?" It is trained always to produce the most likely continuation, never to judge whether it should produce one at all.

1.3 本文做什么、不做什么

1.3 What this paper does and does not do

本文**不**主张 AI 有意识、有自我、有主观体验。校准、拒答、可辨识性自知全都是行为层面的工程量，一台机器可以在"毫无体验"的情况下完成它们（第 6 章会正面处理这个反驳）。

This paper does **not** claim that AI has consciousness, a self, or subjective experience. Calibration, abstention, and identifiability awareness are all behavioral, engineering quantities; a machine can perform all of them with no inner experience whatsoever (Section 6 faces this objection head-on).

本文**要**做的，是给"自知"找到三个工程必要条件，把它们做成可测量、可验收、可证伪的对象。打个比方：我们不是在造发动机（能力由结构与规模决定，那是 P0/P0b 的主题），我们是在装**刹车**。刹车不产生动力，但一辆没有刹车的车，马力越大越危险。

What the paper **does** is find three engineering prerequisites for "self-knowledge" and make them measurable, acceptance-tested, and falsifiable. We are not building the engine — ability is the subject of P0/P0b — we are installing the **brakes**. Brakes produce no power, but a brakeless car becomes more dangerous the more powerful its engine is.

2 把"自知"拆成三个能测量的东西

2 Taking "Self-Knowledge" Apart into Three Measurable Things

"知道自己知道什么"听起来很玄，拆开之后每一部分都有成熟的数学工具。

"Knowing what you know" sounds mystical; taken apart, each piece has mature mathematical tooling.

2.1 校准：说八成把握，就该对八成

2.1 Calibration: 80% confident should mean 80% correct

校准 (calibration) 衡量置信度与正确率是否一致。如果一个模型对 100 道题都说"我 90% 确定"，最后恰好对了 90 道，它就是完美校准的；如果只对了 55 道，它就是过度自信。

Calibration asks whether stated confidence matches actual correctness. A model that says "90% confident" on 100 questions and gets exactly 90 right is perfectly calibrated; if it gets 55 right, it is overconfident.

常用指标是期望校准误差 ECE（把样本按置信度分桶，比较每桶的置信度均值与正确率均值之差）和 Brier 分数（置信度与对错的均方误差）。**关键点：校准和准确率是两件事。** 一个准确率 70% 的模型可以校准得很好（它知道自己只有七成把握），一个准确率 95% 的模型也可能校准得很差（它总说自己百分百对）。Guo 等（2017）发现，现代深度神经网络普遍过度自信，而一个叫"温度缩放"的单参数后处理，能显著改善校准却几乎不改变准确率——这说明"知道自己对不对"和"对不对"是可以分离的两个轴。

Standard tools are the Expected Calibration Error (ECE — bin samples by confidence and compare each bin's mean confidence with its accuracy) and the Brier score (mean squared error between confidence and correctness). **The key point: calibration and accuracy are different things.** A 70%-accurate model can be well calibrated (it knows it is right only seven times in ten), while a 95%-accurate model can be badly calibrated (it always claims certainty). Guo et al. (2017) found that modern deep networks are systematically overconfident, and that a one-parameter post-processing step called temperature scaling substantially improves calibration with almost no change in accuracy — evidence that "knowing whether you are right" and "being right" are separable axes.

2.2 选择性拒答：允许说"我不答"

2.2 Selective abstention: permitting the words "I won't answer"

选择性预测 (selective classification) 给模型加一个弃权开关：对置信度低于阈值的样本，模型不回答，转交给人类或其他系统。

Selective classification adds an abstention switch: on inputs whose confidence falls below a threshold, the model does not answer and hands off to a human or another system.

它的核心工具是**风险—覆盖率曲线 (risk-coverage curve)**：覆盖率是模型答了多少比例的样本，风险是它在已答样本上的错误率。把置信度阈值从松调到紧，覆盖率下降、风险也下降，整条曲线越靠左下越好。一个汇总指标是曲线下面积 AURC，**越小越好**。这个框架最早可追溯到 Chow（1957）的最优误差—拒绝权衡，经 El-Yaniv 与 Wiener（2010）、Geifman 与 El-Yaniv（2017）形式化。

Its central tool is the **risk-coverage curve**: coverage is the fraction of samples the model answers; risk is its error rate on the answered samples. Sweeping the confidence threshold from loose to tight lowers both

coverage and risk, and the curve should sit as far toward the lower-left as possible. A scalar summary is the area under this curve, AURC, where **smaller is better**. The framework goes back to Chow's (1957) optimum error-reject trade-off and was formalized by El-Yaniv & Wiener (2010) and Geifman & El-Yaniv (2017).

拒答不是越多越好——在医疗、自动驾驶场景，“我不知道”有时比答错更危险（它把责任推回给了已经分心的人类）。好的系统让使用者在风险预算下显式权衡，而不是简单地“不确定就拒”。

More abstention is not always better — in medicine or autonomous driving, "I don't know" can be more dangerous than a wrong answer when it hands control back to an already-overloaded human. A good system lets the operator make an explicit trade-off under a risk budget rather than simply refusing whenever uncertain.

2.3 可辨识性自知：有些东西在当前观测下根本不可知

2.3 Identifiability awareness: some things are unknowable under the current observations

前两个量都在问“我对这个答案有多确定”。第三个量更根本：**这个量在当前观测结构下，到底能不能被估计出来？**

The first two quantities ask "how sure am I of this answer?" The third is more fundamental: **can this quantity be estimated at all under the current observation structure?**

系统辨识理论区分两个概念（Bellman & Åström, 1970）：

Systems identification (Bellman & Åström, 1970) distinguishes:

- **可观测性 (observability)**：从一段观测轨迹，能否恢复系统的内部状态？它是一个“是/否”的结构属性；
- **可辨识性 (identifiability)**：给定数据，能否把未知参数估计到足够精度？它是一个“能估到多准”的问题。

- **Observability**: can the internal state be recovered from an observation trajectory? A yes/no structural property.
- **Identifiability**: given the data, can the unknown parameters be estimated accurately enough? A question of "how well."

一个量可以高度可观测（数据里确实有它的影子），却仍然不可辨识（影子太弱、样本太少，估计误差大过不用模型）。**有自知的系统必须能识别这种情况并拒绝硬猜**——这是本文相对传统校准文献最想补上的一块。我们用一个“自知分数”来量化：

A quantity can be highly observable (its shadow is present in the data) yet remain non-identifiable (the shadow is too weak or the sample too short, so the estimator is worse than not modeling at all). **A self-aware system must detect this and refuse to guess** — the piece this paper most wants to add to the classical calibration literature. We quantify it with a skill score:

$$\text{skill} = 1 - \frac{\text{估计器误差}}{\text{直接报平均值的误差}} \tag{1}$$

分数为正，说明模型比"无脑报均值"强；分数为零，说明模型白学；**分数为负，说明模型比什么都不做还差——这时最理性的动作就是别用它。**

A positive score means the model beats the naive mean; zero means it learned nothing; **a negative score means the model is worse than doing nothing — and the rational action is not to use it.**

2.4 三元组：一个工程化的最小框架

2.4 The triad: a minimal engineering framework

把三件事合起来，本文用一个三元组描述"工程意义上的自我模型"：

Putting the three together, we describe an "engineering self-model" as a triad:

$\big(M_{\text{self}}, U, A(x, \tau)\big)$

- M_{self} （自我模型）：系统对自身能力的估计——我在这类任务上历史表现如何；
- U （不确定性）：对当前预测给出的不确定度，且要区分两种来源——**偶然不确定性**（数据本身的噪声，给再多数据也消不掉）与**认知不确定性**（因为没见过、数据不够，可以靠补数据消除）；
- $A(x, \tau)$ （拒答策略）：当不确定度超过阈值 τ 时弃权、转人工或调用工具。

- M_{self} (self-model): the system's estimate of its own competence — how it has historically performed on this kind of task;
- U (uncertainty): the uncertainty attached to the current prediction, distinguishing **aleatoric uncertainty** (noise intrinsic to the data, irreducible with more data) from **epistemic uncertainty** (uncertainty from not having seen enough, removable by collecting more data);
- $A(x, \tau)$ (abstention policy): abstain, hand off, or call a tool when uncertainty exceeds threshold τ .

不确定性还要能给出**有覆盖保证的区间**而不只是一个点，这由共形预测提供（第 4.2 节）。

Uncertainty must also come with **coverage-guaranteed intervals**, not just points; conformal prediction provides them (Section 4.2).

顺带说明本文的层次定位。认知可以粗分四层：L1 感知（识别模式）、L2 认知（推理规划）、L3 元认知（知道自己知道什么、不知道什么）、L4 超认知（对"自己的校准"再校准，乃至主动提出新问题）。本文只做 L3，L4 是开放问题。这个四层划分是本文的组织框架，不是文献共识。

A note on levels. Cognition can be coarsely layered: L1 perception (pattern recognition), L2 cognition (reasoning and planning), L3 metacognition (knowing what you do and do not know), L4 hyper-cognition (recalibrating your own calibration, even posing new questions). This paper does L3 only; L4 is open. The four-layer scheme is our organizing frame, not a consensus of the literature.

图1 元认知闭环

图 1：元认知闭环示意。系统行动后观测结果，世界模型给出预测与不确定度，校准/可辨识性模块判断"我知不知道"，拒答策略决定作答、弃权还是转外部，并把结果写回自我模型。

Figure 1: The metacognition loop. After acting, the system observes the outcome; the world model produces a prediction and an uncertainty; calibration/identifiability modules judge "do I know this?"; the abstention policy decides whether to answer, refuse, or go external; the result is written back into the self-model.

3 前人已经走过的路

3 The Road Already Traveled

本文不是凭空起楼，四条研究脉络在过去几十年分别把三块拼图磨了出来。

This paper is not built in a vacuum. Four research lines have, over decades, shaped the three pieces.

元认知概念来自发展心理学。Flavell（1979）提出"元认知"一词，把它分为元认知知识（对自己认知过程的了解）与元认知监控（对自己当前理解状态的监视）。本文借用的是后者的工程对应物：系统对自身预测可靠性的在线监视。

The concept of metacognition comes from developmental psychology. Flavell (1979) coined the term and divided metacognition into metacognitive knowledge (understanding one's own cognitive processes) and metacognitive monitoring (watching one's current state of comprehension). We borrow the engineering counterpart of the latter: online monitoring of the reliability of one's own predictions.

校准在深度学习时代被重新发现。Guo 等（2017）证明现代网络过度自信，温度缩放（把 logits 除以一个标量 T 再做 softmax）能以极小代价改善校准、且不改变预测类别——这直接支撑本文的预测 F1。

Calibration was rediscovered in the deep-learning era. Guo et al. (2017) showed modern networks are overconfident and that temperature scaling (dividing logits by a scalar T before softmax) improves calibration cheaply without changing predicted classes — direct support for prediction F1.

拒答有一条完整的理论线：Chow（1957）提出带拒绝选项的最优识别；El-Yaniv 与 Wiener（2010）建立无噪声选择性分类的风险—覆盖框架；Geifman 与 El-Yaniv（2017）给出任意训练好网络的选择性分类器；SelectiveNet（2019）把拒绝选项直接做进网络结构。

Abstention has a complete theoretical line: Chow (1957) introduced optimal recognition with a reject option; El-Yaniv & Wiener (2010) founded the risk–coverage framework for noise-free selective classification; Geifman & El-Yaniv (2017) built a selective classifier for any trained network; SelectiveNet (2019) baked the reject option into the network itself.

分布无关的不确定性量化由共形预测提供。Angelopoulos 与 Bates（2021）的导论系统介绍了分裂共形法：在一批没参与训练的校准数据上算分位数，就能给预测区间一个不依赖任何分布假设、只依赖"校准样本与测试样本可交换"的覆盖率保证。

Distribution-free uncertainty quantification is supplied by conformal prediction. The Angelopoulos & Bates (2021) primer systematizes split conformal: compute a quantile on calibration data untouched by

training, and prediction intervals inherit a coverage guarantee that assumes nothing about the distribution — only exchangeability between calibration and test samples.

大模型到底知不知道自己知道什么？ Kadavath 等（2022）做了关键实证：大模型在多项选择里"非语言地"预测自己对错的能力，与真实正确率相关但远不完美，而且 RLHF 微调会损害校准。Reflexion (Shinn 等, 2023) 让智能体把反思写进语言记忆。世界模型 (Ha & Schmidhuber, 2018) 则提示了另一条路：用预测误差本身作为"我没见过这个"的信号。Hendrycks 与 Gimpel (2017) 发现最大 softmax 值可作为检测分布外样本的廉价基线。

Do large models know what they know? Kadavath et al. (2022) ran the key experiment: models' non-verbal ability to predict their own correctness in multiple-choice settings correlates with actual accuracy but is far from perfect, and RLHF fine-tuning harms calibration. Reflexion (Shinn et al., 2023) has agents write reflections into language memory. World Models (Ha & Schmidhuber, 2018) suggests another route: use prediction error itself as the signal for "I have not seen this." Hendrycks & Gimpel (2017) found the maximum softmax value to be a cheap baseline for out-of-distribution detection.

这些工作合起来留下一个缺口：**校准文献很少问"这个量在结构上可不可估"，可辨识性文献很少接大模型的拒答工程，共形预测很少被放进一个统一的"元认知"框架里。** 本文的工作就是把三块拼起来，并在一个可控的小世界模型上端到端跑通。

Together these leave a gap: **the calibration literature rarely asks whether a quantity is structurally estimable; the identifiability literature rarely connects to LLM abstention engineering; conformal prediction is rarely placed inside a unified "metacognition" frame.** This paper joins the three pieces and runs them end-to-end on a controllable small world model.

4 我们实测到了什么

4 What We Actually Measured

实验平台是 UDOS 引擎里的小型世界模型：2 层门控循环网络，隐藏维 256，场景潜变量 32 维，输入 6 帧窗口，共 **967,796 个参数**。它在合成的二维动力学世界里学习"状态—动作—下一状态"的转移。这个世界有意被设计成包含四类动力学：匀速、加速度、弹簧、碰撞。

The platform is a small world model inside the UDOS engine: a 2-layer gated recurrent network, hidden size 256, 32-dimensional scene latent, 6-frame input windows, **967,796 parameters** in total. It learns the state-action-next-state transition in a synthetic 2-D dynamical world deliberately containing four regimes: uniform motion, acceleration, springs, and collisions.

4.1 世界模型本身：先证明它能预测

4.1 The world model itself: first prove it can predict

从 v7.0.1 到 v7.0.3，盲测集上的整体预测误差 (MSE) 从 0.0459 降到 **0.01037**，约改善 4.4 倍。分类别看，碰撞最容易 (0.00304)、匀速次之 (0.00573)、加速度中等 (0.01410)、弹簧最难 (0.01862)。单步预测 (批大小 1) 中位延迟 2.36 毫秒，4 步推演 4.83 毫秒——轻到可以放进实时控制回路。(等级 verified)

From v7.0.1 to v7.0.3, overall prediction error (MSE) on the blind test set fell from 0.0459 to **0.01037**, roughly a $4.4\times$ improvement. By regime, collisions are easiest (0.00304), then uniform motion (0.00573), acceleration (0.01410), and springs hardest (0.01862). Median single-step prediction latency (batch size 1) is 2.36 ms; a 4-step rollout takes 4.83 ms — light enough for a real-time control loop. (Level verified.)

这一步是后续一切的前提：一个连预测都做不好的模型，谈不上评估自己预测得好不好。

This step is the precondition for everything else: **a model that cannot predict cannot assess how good its predictions are.**

4.2 共形预测：承诺 95%，实测 95.8%

4.2 Conformal prediction: promise 95%, observe 95.8%

我们用分裂共形法给世界模型的预测加区间：640 个窗口、128 条轨迹作为留出校准集，然后在全新测试轨迹上检查“真实值落进区间”的比例。分两条路径：oracle 路径允许看到隐藏物理参数的真值，blind 路径完全看不到（更接近真实部署）。

We wrap the world model's predictions in split-conformal intervals: 640 windows and 128 trajectories serve as a held-out calibration set; on fresh test trajectories we check how often the true value falls inside. Two paths are run: an oracle path allowed to see the true latent parameters, and a blind path that cannot (closer to real deployment).

表 1 共形覆盖率：承诺值 vs 实测值（等级 verified， ± 0.05 容差）

Table 1 Conformal coverage: nominal vs observed (level verified, ± 0.05 tolerance)

名义水平	oracle 实测覆盖	oracle 区间带宽	blind 实测覆盖	blind 区间带宽
80%	0.7839	0.0430	0.7758	0.0504
90%	0.8984	0.0858	0.9001	0.1081
95%	0.9577	0.1612	0.9559	0.2052

English table below / 英文表格如下：

Nominal	Oracle coverage	Oracle width	Blind coverage	Blind width
80%	0.7839	0.0430	0.7758	0.0504
90%	0.8984	0.0858	0.9001	0.1081
95%	0.9577	0.1612	0.9559	0.2052

图2 共形覆盖率

图 2：三个名义水平下，实测覆盖率都落在 ± 0.05 容差内（21 项门控检查全部通过）。blind 路径看不到隐藏参数，覆盖率依然达标，代价是区间系统性地更宽（95% 水平带宽 0.2052 vs 0.1612）。

Figure 2: At all three nominal levels, observed coverage lies within ± 0.05 (all 21 release gates pass). The blind path, which cannot see latent parameters, still meets coverage, at the price of systematically wider intervals (0.2052 vs 0.1612 at 95%).

怎么读这张表：模型不需要"感觉"自己有多确定，它只需要在一批校准数据上老实地量一次分位数，就能给出有数学保证的区间——"我说 95%，长期来看就真的有 95% 左右的真实值落进来"。blind 路径更宽的带宽尤其重要：**系统在信息更少时，自动变得更"谦虚"，这正是边界感知的雏形。**

How to read this table. The model need not "feel" how certain it is; it simply measures a quantile once on calibration data and obtains intervals with a mathematical guarantee — "when I say 95%, roughly 95% of future true values fall in." The wider blind intervals matter most: **with less information the system automatically becomes more humble — the germ of boundary awareness.**

共形保证有一个明确前提：校准数据与测试数据可交换。分布漂移、分布外场景下这个前提失效，覆盖率会衰减——这是本文预测 F4 要处理的问题，我们不假装它不存在。

The conformal guarantee has an explicit premise: calibration and test data must be exchangeable. Under distribution shift or out-of-distribution inputs the premise fails and coverage degrades — the subject of prediction F4, not a limitation we paper over.

4.3 加速度负结果：当"硬猜"比"不猜"更糟

4.3 The acceleration negative result: when guessing is worse than not guessing

这是全文最反直觉、也最有价值的一组结果。我们让估计器从 6 帧观测窗口里反推四个隐藏物理量，每个量 160 个样本，用式（1）算自知分数：

This is the most counterintuitive and valuable result. We ask the estimator to recover four latent physical quantities from a 6-frame window, 160 samples each, and compute the skill score of Eq. (1):

表 2 四个隐参数的可辨识性 (n=480, 主门 verified, 细分为 cpu-proto)

Table 2 Identifiability of four latent quantities (n=480; main gate verified, breakdown cpu-proto)

隐参数	估计器误差	报均值的误差	自知分数 skill	可观测性
初始速度 v_0	0.6606	0.9683	+0.318	0.771
角频率 ω	0.1471	0.2563	+0.426	0.914
碰撞弹性 v_2	0.0432	0.2396	+0.820	0.908
加速度 a (6 帧窗口)	0.9256	0.6138	-0.508	0.798

English table below / 英文表格如下：

Latent quantity	Estimator error	Mean-prediction error	Skill	Observability
Initial velocity v_0	0.6606	0.9683	+0.318	0.771
Angular frequency ω	0.1471	0.2563	+0.426	0.914
Collision restitution v_2	0.0432	0.2396	+0.820	0.908
Acceleration a (6-frame window)	0.9256	0.6138	-0.508	0.798

图4 可辨识性自知

图 4：前三个参数的自知分数为正，模型确实学到了东西；加速度在 6 帧窗口下为 -0.508，估计器比直接报平均值还差 50.8%。注意它的可观性并不低（0.798）——数据里有加速度的影子，但 6 帧太短，不足以把它辨识出来。

Figure 4: The first three quantities have positive skill — the model genuinely learns them. Acceleration under a 6-frame window scores -0.508: the estimator is 50.8% worse than predicting the mean. Note that its observability is not low (0.798) — acceleration's shadow is in the data, but six frames are too few to identify it.

这个负结果说明三件事。 第一，可观性高不等于可辨识：加速度在数据里有影子，但窗口太短，任何模型都估不准。第二，一个没有自知的系统此时会自信地输出那个糟糕的点估计——这就是一次教科书式的幻觉；一个有自知的系统看到 skill 为负，应当弃权或把观测窗口从 6 帧加长到 12/24 帧。第三，也是预测 F2 的核心：**这是观测结构的硬边界，不是靠把模型做大就能绕过去的。**

The negative result says three things. First, high observability does not imply identifiability: the shadow of acceleration is present, but the window is too short for any model to estimate it. Second, a system without self-awareness would here confidently emit that terrible point estimate — a textbook hallucination; a self-aware system that sees negative skill should abstain or extend the window from 6 to 12/24 frames. Third — and this is the core of prediction F2 — **this is a hard boundary of the observation structure, not something a bigger model routes around.**

4.4 门控：把"不知道"接到真实动作上

4.4 Gating: wiring "I don't know" into real actions

校准和区间如果只停留在报表里，就不产生安全价值。我们把不确定度接成了一组发布前门控（gating），共 21 项检查，全部通过（覆盖率容差 ± 0.05 、参数扇形 ± 0.10 ）。

Calibration and intervals produce no safety value if they stay in reports. We wire uncertainty into a set of pre-release gates, 21 checks in total, all passing (coverage tolerance ± 0.05 , parameter fan ± 0.10).

两个最能说明问题的门控：

Two gates make the point most clearly:

- **接触门控：**在具身混合控制实验里（每臂 40 个回合，等级 cpu-proxy），纯语义层直接控制机械臂，成功率 75%、碰撞率 0.25、平均接管 1.0 次；只靠动作先验运动控制，成功率 52.5%；而在接触类任务上，没有动作先验的直接控制成功率是 **0%**，门控识别后把这类任务全部切给混合策略，最终混合策略成

功率 100%、碰撞率 0、平均接管仅 0.047 次。门控不是让模型变聪明，而是阻止它在不擅长的场景里硬上。

- **熔断开关**：当重试预算被设为 0 时，系统如实拒绝接受任务、治理检查返回不通过并降级——它没有为了"完成任务"而强行通过。

- **Contact gate**: in embodied hybrid-control experiments (40 episodes per arm, level cpu-proxy), pure semantic control of the arm succeeds 75% with collision rate 0.25 and 1.0 takeover on average; motion-prior-only control succeeds 52.5%; on contact tasks, direct control without a motion prior succeeds 0% of the time. Once the gate routes contact tasks to the hybrid policy, it reaches 100% success, zero collisions, and only 0.047 takeovers. **The gate does not make the model smarter; it stops it from forcing its way through tasks it is bad at.**
- **Kill switch**: when the retry budget is set to zero, the system honestly refuses the task, the governance check returns not-ok and it degrades — it does not fake completion to "finish the job."

门控的触发率本身也携带信息：匀速场景触发 63.4%、加速度 66.2%、碰撞 15.4%、弹簧 0%。模型在自己更难辨识的动力学上更频繁地求助，这与第 4.3 节的可辨识性排序一致。

Gate trigger rates carry information too: uniform motion 63.4%, acceleration 66.2%, collisions 15.4%, springs 0%. The model calls for help more often on the dynamics it finds harder to identify, consistent with the ordering in Section 4.3.

4.5 群体层：一个智能体的刹车，和一百万个智能体的刹车

4.5 The collective level: brakes for one agent and for a million

自我边界在多智能体系统里还有一层集体含义：整个军团也要知道"自己还能不能收口"。三个来自姊妹论文的结果：

In multi-agent systems the self-boundary has a collective meaning: the whole swarm must also know whether it can still converge. Three results from companion papers:

- 拜占庭容错委员会在 $n=7, f=2$ 配置下对 24 个工单全部裁决正确（法定人数 $2f+1$ ）；
 - 压力注入实验中，带 Owner/Trace/停止条件治理矩阵的系统在正常与加压两种条件下都 24/24 收口，并通过 5 次返工隔离了 3 个故障智能体（**单种子试点，需 ≥ 30 种子**）；
 - 内部任务市场的预算、支付、余额严格守恒（ $240=240=240$ ），账目守恒本身就是一个"系统状态是否仍然自洽"的全局自检信号。
- A Byzantine-fault-tolerant committee with $n=7, f=2$ decides all 24 work orders correctly (quorum $2f+1$);
 - Under fault injection, a system with the Owner/Trace/Stop-Condition governance matrix converges 24/24 in both clean and stressed runs, and isolates 3 faulty agents through 5 reworks (**single-seed pilot; ≥ 30 seeds required**);
 - The internal task market conserves budget, payment, and balance exactly ($240=240=240$); conservation is itself a global self-check that the system state remains consistent.

4.6 证据汇总

4.6 Evidence summary

表 3 三组核心证据与元认知三元组的对应

Table 3 Mapping the three core evidence sets to the metacognition triad

三元组组件	实验	关键数字	等级
不确定性 $U(x)$ (校准)	分裂共形	95% 水平实测 0.9577 (oracle) / 0.9559 (blind)	verified
自我模型 M_{self} (可辨识性自知)	四参数辨识	加速度 skill=-0.508, 应弃权	主门 verified
拒答策略 $A(x, \tau)$	21 项门控	全部通过; 接触任务 0%→100%	verified/cpu-proxy

English table below / 英文表格如下:

Triad component	Experiment	Key number	Level
Uncertainty U (calibration)	Split conformal	0.9577 oracle / 0.9559 blind at 95%	verified
Self-model M_{self} (identifiability awareness)	Four-parameter identification	Acceleration skill -0.508 → abstain	main gate verified
Abstention policy $A(x, \tau)$	21 gates	All pass; contact tasks 0%→100%	verified/cpu-proxy

5 怎么证明本文是错的

5 How This Paper Could Be Proved Wrong
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保留五条预测中判决力最强的三条，每条给出最小实验与判据。

We keep the three most decisive of five predictions, each with a minimal experiment and a decision rule.
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F1 | 校准与性能是两个可分离的轴。 最小实验：取一个训好的模型（无需重训），冻结权重，在校准集上对温度 T 做一维网格搜索，5 折交叉测量 ECE 与准确率。判据：若存在某个 T 使 ECE 相对下降 $\geq 30\%$ 而准确率变化在噪声内 ($<0.1\%$)，F1 成立；若校准每次改善都伴随准确率同比例改善，F1 死亡（校准只是性能的附属品）。

F1 | Calibration and performance are separable axes. Minimal experiment: take one trained checkpoint (no retraining), freeze the weights, do a one-dimensional grid search over temperature T on a calibration set, and measure ECE and accuracy with 5-fold cross-validation. Decision rule: F1 holds if some T cuts ECE by $\geq 30\%$ relative while accuracy moves within noise ($<0.1\%$); F1 dies if every measurable calibration gain comes with a proportional accuracy gain (calibration is a mere by-product of performance).

F2 | 观测结构不足时，加参数救不回来。 最小实验：固定 6 帧窗口，只把估计器容量翻倍、再翻四倍（隐藏维 256→512→1024，层数 2→4），测加速度的 skill；同时设正对照臂，把窗口加到 12/24 帧。判据：若加容量后 skill 始终 ≤ 0 、而加长窗口后立刻转正，就干净地分离了"观测结构"与"模型容量"两个变量，F2 成立；若大模型靠容量把 skill 稳定抬到 +0.2 以上，F2 死亡——那意味着规模确实能突破观测结构的硬边界，本文退为"小模型的约束"。这是本文最激进、也最容易被检验的一条。

F2 | When the observation structure is insufficient, more parameters cannot rescue it. Minimal experiment: hold the 6-frame window fixed, only double and quadruple estimator capacity (hidden 256→512→1024, layers 2→4), and measure acceleration skill; run a positive-control arm that extends the window to 12/24 frames. Decision rule: F1 holds if skill stays ≤ 0 as capacity grows but turns positive immediately with a longer window — cleanly separating observation structure from model capacity; F2 dies if a big model lifts skill stably above +0.2, meaning scale does break the structural boundary and the paper retreats to "a small-model constraint." This is the boldest and most easily tested prediction.

F3 | 门控在同样的错误率预算下能答得更多。 最小实验：同一模型设两臂（裸模型 vs 加共形/置信门控），固定目标错误率，扫描阈值，比较覆盖率与 AURC。判据：门控臂的覆盖率显著更高（AURC 显著更小）则 F3 成立；两条风险—覆盖曲线在统计噪声内重合则 F3 死亡（拒答只是把难题推给别人）。

F3 | Gating answers more under the same error budget. Minimal experiment: on one benchmark run two arms — bare model vs the same model plus conformal/confidence gating; fix a target error rate, sweep the threshold, compare coverage and AURC. Decision rule: F3 holds if the gated arm covers significantly more (significantly smaller AURC); F3 dies if the two risk-coverage curves coincide within statistical noise (abstention merely passes hard problems to someone else).

另有两条次级预测：**F4** 显式不确定性通道（共形区间、参数扇形）在分布外数据上应比模型口头自报的置信度（verbalized confidence）退化更慢——Kadavath 已发现口头置信度不完美且会被 RLHF 损害；**F5** 带治理矩阵的多智能体系统在故障注入下收口率更高，但现有 24/24 只是单种子点估计，**必须 ≥ 30 种子、报告差值的 95% 置信区间**，置信区间包含零则 F5 死亡。

Two secondary predictions: **F4**, explicit uncertainty channels (conformal intervals, parameter fans) should degrade more slowly than verbalized confidence under out-of-distribution shift — Kadavath already found verbal confidence imperfect and harmed by RLHF; **F5**, governed multi-agent systems should converge more often under fault injection, but the current 24/24 is a single-seed point estimate and **must be rerun with ≥ 30 seeds with a 95% confidence interval**; if that interval contains zero, F5 dies.

6 我们可能错在哪里

6 Where We Might Be Wrong

第一，校准不是意识。 温度缩放只是除以一个标量，共形预测只是查一次分位数表，门控只是比较两个数的大小——没有任何一步需要系统"反思自己在反思"。我们完全接受这个行为主义反驳，并把它写进标题：本文说的"元认知"是一个工程术语，等于"校准 + 拒答 + 可辨识性自知"这三个行为量，不是 Flavell 意义上的元认知体验，更不是意识。一个系统可以在毫无主观体验的情况下完美校准。把"自知"从内在体验问题降维成行为测量问题，恰恰是本文能把它做出来的前提。

First, calibration is not consciousness. Temperature scaling divides by a scalar; conformal prediction looks up a quantile; a gate compares two numbers — none of this requires the system to "reflect on its own reflection." We fully accept the behaviorist objection and put it in the title: our "metacognition" is an engineering term denoting the three behavioral quantities, not Flavell's metacognitive experience, let alone consciousness. A system can be perfectly calibrated with no subjective experience at all. Reducing self-knowledge from a question of inner experience to one of behavioral measurement is precisely what lets us build it.

第二，过度拒答是另一种失职。 稍微不确定就弃权的系统会失去使用价值；在医生稀缺时把患者推出去、在驾驶员分神时把控制权甩回去，可能比硬答更危险。置信度还可能被对抗样本欺骗，共形的可交换性前提在分布外失效。本文目前的 AURC 只度量了"答了多少、错了多少"，没有度量"拒了多少、拒的代价多大"——拒答成本建模是明确的下一步。

Second, over-abstention is another kind of failure. A system that refuses at the slightest doubt is useless; in a doctor-scarce setting, handing the patient back, or handing control back to a distracted driver, can be worse than answering. Confidence can also be fooled by adversarial inputs, and conformal exchangeability fails out of distribution. Our AURC currently measures only "how much answered vs how much wrong," not "how much refused and at what cost" — modeling the cost of refusal is explicit future work.

第三，内省本身不可靠。 让模型直接说"我有几成把握"，这个自述也需要被校准；Reflexion 式的语言反思还可能把错误原因写进记忆、污染下一次。所以本文的立场是一句话：**不要信它说的，要看它测的。** 模型内部的置信信号可以当候选，但必须用独立校准集把它校准成可靠量。

Third, introspection itself is unreliable. Asking a model to state its own confidence produces a self-report that itself needs calibrating; Reflexion-style verbal reflection can mis-attribute a failure and write the wrong cause into memory, contaminating the next trial. Hence our one-line stance: **do not trust what it says; trust what it measures.** Internal confidence signals may be candidates, but they must be calibrated against an independent calibration set.

第四，孔子不是证据，合成世界不是真实世界。 《论语》只作引言；全部 UDOS 数字都来自 CPU 上的确定性合成原型（点质量动力学、规则裁判、单一种子），不是真实大模型、不是真机、没有 95% 置信区间。加速度的负结果是在人为注入的短窗口缺陷上测的，真实世界的"观测不足"形态更复杂。因此本文把自己定位为**研究纲领加硅基初步证据**：它证明机制的形状在简化世界里跑得通，不声称常数在真实系统里成立。把共形预测搬到真实大模型的输出空间、用 ≥ 30 种子复算全部试点，是投稿前必须补的两件事。

Fourth, Confucius is not evidence, and a synthetic world is not the real world. The Analects only motivates; every UDOS number comes from a deterministic CPU prototype (point-mass dynamics, rule-based judges, a single seed) — not a real large model, not real hardware, with no 95% confidence intervals. The acceleration negative result was measured on an injected short-window defect; real-world "insufficient observation" is more complex. We therefore position this paper as **a research program plus preliminary silicon evidence**: it shows the mechanism's shape works in a simplified world, not that its constants hold in real systems. Moving conformal prediction onto real LLM output spaces and rerunning every pilot with ≥ 30 seeds are the two things that must be done before submission.

7 意识问题：本文明确不碰

7 Consciousness: Explicitly Out of Scope

本章单独声明越界。本文全文**不主张**系统有意识、感受 (qualia) 或自我感。注意力图式理论 (AST)、全局工作空间理论 (GWT) 等意识假说与本文的自我模型有表面相似，但本文**不背书、不引用、不检验**它们：它们是关于人类意识的假说，能否用于硅基系统本身就有争议，而本文的 M_{self} 只是一个对自身能力的数值估计，与"注意力图式"不是同一个概念。

This chapter marks the boundary. The paper nowhere **claims** consciousness, qualia, or a sense of self. Theories such as Attention Schema Theory (AST) or Global Workspace Theory (GWT) superficially resemble our self-model, but we **neither endorse, cite, nor test** them: they are hypotheses about human consciousness whose applicability to silicon is itself contested, while our M_{self} is merely a numeric estimate of the system's own competence — not an "attention schema."

一个校准的系统不必有意识，一个有意识的系统也未必校准。这一章的存在本身就是"知之为知之"纪律的体现：**本文知道自己不知道意识是什么，也不假装知道。**

A calibrated system need not be conscious; a conscious system need not be calibrated. The existence of this chapter is itself the discipline of "knowing what you do not know": **this paper knows it does not know what consciousness is, and does not pretend to.**

8 工程启示：先装刹车，再踩油门

8 Engineering Implications: Install the Brakes Before Flooring the Throttle

其一，校准应当是一等验收指标，而不是上线后的补丁。 温度缩放很便宜，但前提是你得去测它。把 ECE、覆盖率、AURC 写进持续集成的门禁（本文的 21 项门控就是这么做的），比事后救火便宜得多。

First, calibration should be a first-class acceptance criterion, not a post-launch patch. Temperature scaling is cheap, but only if you actually measure it. Writing ECE, coverage, and AURC into CI gates (as our 21 gates do) is far cheaper than firefighting afterward.

其二，幻觉要管理，而不是幻想"消除"。 你不可能让有限系统在所有样本上都对；你能做的是让它在不知道的时候做三件事之一：输出宽区间（共形）、弃权（门控）、转去查外部知识（RAG/Toolformer）。"用更大模型消除幻觉"的思路只对分布内有效——幻觉的根源是自我边界缺失，不是规模不足，这与 P0b 的预测 F2 直接呼应。

Second, hallucination is to be managed, not fantasized away. No finite system is right on every sample; what you can do is ensure that, when it does not know, it does one of three things: emit a wide interval (conformal), abstain (gating), or retrieve external knowledge (RAG/Toolformer). "Eliminate hallucination with a bigger model" works only in-distribution — the root of hallucination is a missing self-boundary, not insufficient scale, in direct correspondence with prediction F2 of P0b.

其三，观测结构要在设计阶段就考虑可辨识性。加速度负结果提醒架构师：如果观测窗口太短、传感器看不到关键量，换再大的模型也没用，正确动作是加长窗口、增加传感器，而不是堆参数。

Third, identifiability must be considered at observation-design time. The acceleration result tells architects: if the window is too short or the sensors cannot see a key quantity, no larger model helps; the correct move is a longer window or better sensors, not more parameters.

其四，三部曲是乘法关系。P0 说结构决定能力函数长什么样，P0b 说规模决定站在曲线哪一段，本文说自我边界决定走到哪里必须停。结构与规模是发动机，自我边界是刹车：一辆刹车失灵的车，动力越强越危险；一辆只有刹车没有动力的车，则只是路障。三者缺一不可。

Fourth, the trilogy is multiplicative. P0 says structure determines the shape of the competence function; P0b says scale determines where you stand on the curve; this paper says the self-boundary determines where you must stop. Structure and scale are the engine; the self-boundary is the brake: a brakeless car is more dangerous the more power it has, while a car with only brakes and no engine is just a roadblock. All three are required.

9 结语

9 Conclusion

P0 说：空间结构，本身就是智能。P0b 说：结构加规模，本身就是智能。本文说：**知之为知之，是智能的第三轴。**

P0 says: space structure is itself intelligence. P0b says: structure plus scale is itself intelligence. This paper says: **knowing what one does not know is the third axis of intelligence.**

我们没有证明任何系统拥有智慧。我们做的事情朴素得多：把"智慧"这个无法直接测量的词，替换成校准、拒答、可辨识性自知三个可以测量的工程必要条件，在一个 96.8 万参数的小世界模型上端到端跑通，并立好五个可以被实验推翻的靶子。

We have not proved that any system possesses wisdom. What we did is far more modest: we replaced the unmeasurable word "wisdom" with three measurable engineering prerequisites — calibration, abstention, identifiability awareness — ran them end-to-end on a 968k-parameter world model, and set up five targets that experiments can knock down.

AI 的成人礼，不会是它答对了所有问题的那一天，而是它第一次诚实地说出"我不知道，让我查一下 / 这超出了我的观测能力"的那一天。《论语》把这个理想讲了两千五百年；本文做的，是把它从道德训诫改造成一道可测量、可验收、可证伪的工程命题。在这三件事被做成之前，"这个系统有智慧"的说法都不严谨——无论它的回答多么流畅。

AI's coming-of-age will not be the day it answers every question. It will be the day it first honestly says, "I don't know — let me look that up," or "this lies beyond what my observations can determine." The Analects held that ideal for 2,500 years; what this paper does is convert a moral precept into a measurable,

acceptance-testable, falsifiable engineering proposition. Until the three things are built, calling a system "wise" is not rigorous — however fluent its answers.

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附录：本文 UDOS 数字溯源表

Appendix: Provenance of the UDOS Numbers

数字	数据文件	等级
世界模型 967,796 参数；盲测 MSE 0.0459→0.01037；单步延迟 2.36ms	v7_verification.json (params/improvement_vs_v701/latency_ms)	verified
共形 80/90/95: oracle 0.7839/0.8984/0.9577, blind 0.7758/0.9001/0.9559；带宽见表 1	v7_verification.json (coverage.*)	verified
skill: v0 +0.318、 ω +0.426、v2 +0.820、加速度 -0.508 (n=480)	v7_verification.json (estimator_param_error.*)	主门 verified, 细分 cpu-proto
21 项门控全过；触发率 0.6341/0.6624/0.1544/0	v7_verification.json (gating)	verified
具身混合 direct 75%/motion 52.5%/hybrid 100%，碰撞 0.25→0，接管 1.0→0.047	embodied_hybrid_demo.json (P7)	cpu-proxy
BFT n=7,f=2 全对；市场守恒 240；P3 24/24 (单种子)	P2/P3/P11 回源	cpu-proto

English table below / 英文表格如下：

Number	Data file	Level
World model 967,796 params; blind MSE 0.0459→0.01037; latency 2.36 ms	v7_verification.json (params / improvement_vs_v701 / latency_ms)	verified
Conformal 80/90/95: oracle 0.7839/0.8984/0.9577, blind 0.7758/0.9001/0.9559; widths in Table 1	v7_verification.json (coverage.*)	verified
Skill: v0 +0.318, ω +0.426, v2 +0.820, acceleration -0.508 (n=480)	v7_verification.json (estimator_param_error.*)	main gate verified, breakdown cpu-proto
21 gates all pass; trigger rates 0.6341/0.6624/0.1544/0	v7_verification.json (gating)	verified
Embodied hybrid direct 75%/motion 52.5%/hybrid 100%; collisions 0.25→0; takeovers 1.0→0.047	embodied_hybrid_demo.json (P7)	cpu-proxy
BFT n=7,f=2 all correct; market conservation 240; P3 24/24 (single seed)	P2/P3/P11 sources	cpu-proto

(全文完。P0c 为旗舰三部曲序章"自我边界"轴；结构轴见 P0，规模轴见 P0b。)

(End. P0c is the prologue "self-boundary" axis of the flagship trilogy; the structure axis is P0, the scale axis P0b.)